

Periods, Obstructions, and Duality: A Cohomological Reading of the Geometry of Model Spaces

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June 11, 2026

Abstract

We propose a systematic reading of statistical inference through the de Rham cohomology of the parameter space. After endowing a family of models with the Riemannian structure induced by the Fisher information metric, we show that (i) the Fisher curvature determines, via Chern–Weil theory, canonical representatives of the characteristic classes of M , invariant under reparametrization; (ii) nontrivial classes in $H_{\text{dR}}^1(M)$ obstruct the existence of global potentials for optimization dynamics and manifest themselves as monodromy of the estimator; (iii) loss functions admit a natural promotion from scalars to densities integrable over chains, under which model selection is reinterpreted as a Poincaré duality pairing between loss cocycles and data cycles. We conclude with a conjecture on topological invariance as a regularizing principle against overfitting. The text is deliberately speculative in its extrapolations but rigorous in its foundations: every analogy is anchored in a theorem, and every conjecture is stated so as to be falsifiable.

1 The parameter space as a manifold: Fisher curvature and characteristic classes

Let $\mathcal{P} = \{p_\theta(x) d\mu(x) : \theta \in \Theta\}$ be a statistical family parametrized by an open set $\Theta \subseteq \mathbb{R}^d$, under the usual regularity conditions (common support, differentiation under the integral sign, local identifiability). The correct object is not Θ but the differentiable manifold M of the *distributions themselves*, of which θ is merely a chart. On M lives the Fisher information metric,

$$g_{ij}(\theta) = \mathbb{E}_{p_\theta}[\partial_i \log p_\theta \partial_j \log p_\theta] = -\mathbb{E}_{p_\theta}[\partial_i \partial_j \log p_\theta],$$

the unique metric tensor (up to scale) invariant under sufficient statistics (Chentsov’s theorem). The question organizing this section is: *what information about M does the curvature of g encode that does not depend on g at all?* The answer is precisely the content of Chern–Weil theory, and it is there that de Rham cohomology enters through the front door.

Let ∇ be the Levi-Civita connection of the Fisher metric, with curvature tensor $R \in \Omega^2(M; \text{End}(TM))$. For each invariant polynomial $P \in \mathbb{R}[\mathfrak{gl}_d]^{GL_d}$ of degree k , the form

$$P\left(\frac{i}{2\pi}R\right) \in \Omega^{2k}(M)$$

is *closed*, and its class in $H_{\text{dR}}^{2k}(M)$ is independent of the chosen connection. In particular, the Pontryagin classes $p_k(TM)$ admit representatives built entirely out of the Fisher curvature. The statistical consequence deserves to be stated:

Proposition 1.1 (Inferential invariance of characteristic classes). *Let M be the manifold of a regular statistical family. The Chern–Weil representatives of $p_k(TM)$ constructed from the Fisher curvature change by an exact form under: (a) any diffeomorphic reparametrization of Θ ; (b) any smooth deformation of the family preserving the diffeomorphism type of M ; (c) the replacement of the Levi-Civita connection by any of Amari’s α -connections $\nabla^{(\alpha)}$. The classes $[P(R)] \in H_{\text{dR}}^{2k}(M)$ are therefore invariants of the family, not of the particular parametric model.*

Item (c) is the most interesting one. Information geometry does not work with a single connection but with the one-parameter pencil $\{\nabla^{(\alpha)}\}_{\alpha \in \mathbb{R}}$, where $\nabla^{(1)}$ (exponential) and $\nabla^{(-1)}$ (mixture) are dual with respect to g . The curvatures $R^{(\alpha)}$ vary with α — indeed, in an exponential family $R^{(\pm 1)} \equiv 0$ even though $R^{(0)} \neq 0$ in general — but the *classes* $[P(R^{(\alpha)})]$ do not. Cohomology detects exactly that part of Fisher geometry which is insensitive to the choice of inferential structure.

Example 1.2 (Exponential families: topological triviality). If $\mathcal{P}$ is an exponential family with natural parameter ranging over an open convex set $\Theta \subseteq \mathbb{R}^d$, then $M \cong \Theta$ is contractible and $H_{\text{dR}}^k(M) = 0$ for $k > 0$. Amari’s dual flatness ($R^{(\pm 1)} = 0$) is thus *consistent* with the topology: the characteristic classes have nowhere to live. The moral is that cohomological richness requires leaving the exponential paradise.

Example 1.3 (Mixtures and quotients: where topology is born). Consider a mixture of K Gaussian components. The likelihood is invariant under the action of the symmetric group S_K permuting the component labels (*label switching*). The honest model space is not Θ but the quotient $M = \Theta_0/S_K$, where Θ_0 is the locus on which the action is free. This quotient has nontrivial fundamental group and, generically, $H_{\text{dR}}^1(M) \neq 0$ after passing to suitable coefficients or to strata of the quotient. Analogously, in neural networks the invariance under permutation of hidden neurons and under positive rescalings (for homogeneous activations) produces an effective function space $M = \Theta/G$ with genuinely nontrivial topology. The topology of learning is not a pathology: it is the price of quotienting by the symmetries of the model. Where, in addition, the Fisher metric degenerates (singular models, in Watanabe’s sense), M fails even to be a manifold and the correct tool is the cohomology of stratified spaces; we shall not pursue that generalization here, but it is worth knowing that the door exists.

2 Topological obstructions to optimization: closed non-exact forms

Gradient descent is, in its geometric skeleton, the flow of the field $-\text{grad}_g L$ for a loss $L : M \rightarrow \mathbb{R}$. The associated 1-form, $\omega = \text{d}L$, is exact *by construction*, and there the whole cohomological story would seem to die: the local Poincaré lemma globalizes for free because the potential L exists in advance. But this exactness is an artifact of two assumptions that modern learning violates systematically.

First violated assumption: the dynamics is a gradient. In coupled optimization games (generative adversarial networks, multi-agent learning, certain actor–critic schemes), the update follows a vector field V that is *not* the gradient of any function. Let $\omega = g(V, \cdot)$ be its dual 1-form. If M is compact (or oriented with suitable decay conditions), the Hodge decomposition gives

$$\omega = \text{d}\varphi + \delta\beta + \eta, \quad \eta \in \mathcal{H}^1(M) \cong H_{\text{dR}}^1(M),$$

with η harmonic. The term $d\varphi$ is the “honest” descent component: it admits a Lyapunov function. The term $\delta\beta$ generates local rotation: cycles that averaging or regularization can dissipate. But the harmonic component η is *irreducible*: it represents a nontrivial class in $H_{\text{dR}}^1(M)$, admits no potential even after small perturbations, and its integral curves may wind around non-contractible cycles of M indefinitely. The non-convergence of certain adversarial dynamics is then not a defect of the learning rate or of stochasticity: it is the dynamical shadow of a cohomology class.

Proposition 2.1 (Period obstruction). *Let V be an update field on M whose dual 1-form ω is closed ($d\omega = 0$). There exists a global Lyapunov function L with $dL = \omega$ if and only if all periods vanish:*

$$\oint_{\gamma} \omega = 0 \quad \text{for every } [\gamma] \in H_1(M; \mathbb{Z}).$$

Otherwise, the flow of $-V$ admits recurrent trajectories winding around the cycles γ with $\oint_{\gamma} \omega \neq 0$, and no descent-type algorithm on M following ω can be globally convergent.

The proof is de Rham’s theorem read backwards: a closed ω is exact if and only if its class vanishes, and the class is detected by evaluation against homology. What the proposition adds is the dynamical translation: *nonzero periods are drift rates per loop*. Each circuit around a non-contractible cycle “lowers” the multivalued pseudo-potential by a fixed amount $\oint_{\gamma} \omega$, exactly as the argument $\arg(z)$ drops by 2π per loop around the origin — the canonical example $\omega = d\theta$ on $\mathbb{R}^2 \setminus \{0\}$, closed, non-exact, generator of $H_{\text{dR}}^1()$.

Second violated assumption: the space is that of parameters rather than that of functions. Even when L is a genuine loss on Θ , the invariance $L(\theta) = L(\sigma \cdot \theta)$ under the symmetry group G of the model implies that the relevant dynamics lives on $M = \Theta/G$. The form dL descends to the quotient, but the *estimator* — understood as the assignment data $\mapsto$ critical point reached — acquires monodromy: two SGD trajectories with different seeds or mini-batches may converge to representatives differing by a non-contractible loop in M . For the loss this is invisible; for any functional of the parameter that is not G -invariant (interpretability of individual neurons, coordinatewise confidence intervals, weight averaging across runs) it is fatal.

Remark 2.2 (Consistency with monodromy). Classical asymptotic theory (consistency and asymptotic normality of the MLE) is local: it presupposes a contractible neighborhood of the true parameter θ_0 in which the Poincaré lemma rules without friction. On an M with $\pi_1(M) \neq 0$, consistency must be stated in the quotient: $\hat{\theta}_n \rightarrow \theta_0$ in M , not in Θ . The empirical phenomenon of *mode connectivity* — apparently distinct minima of deep networks joined by paths of nearly constant loss — suggests that much of the multimodality observed in Θ is homotopically trivial in M : distinct modes are lifts of a single point, or points in a single connected component of a sublevel set. The right question is not “how many minima are there?” but “what is the topology of the sublevel set $M_{\leq c} = L^{-1}(-\infty, c]$ near the optimal value?”. The Betti numbers of $M_{\leq c}$, not the count of critical points, are the invariant that Morse theory puts on the table: the Morse inequalities $\#\{\text{critical points of index } k\} \geq b_k(M_{\leq c})$ bound the *unavoidable* complexity of the landscape by the cohomology of the space, regardless of the optimizer’s cleverness.

3 Losses as densities: integration over chains and Poincaré duality

Statistical practice treats the loss as a scalar: $L(\theta) = \mathbb{E}_{x \sim \mathcal{D}}[\ell(x, \theta)]$. We propose to take seriously the fact that this expectation is an *integral*, and that every honest integral is the pairing of a form with a chain.

Let X be the data manifold (the effective support of $\mathcal{D}$, which the manifold hypothesis posits to be of dimension $n \ll$ the ambient dimension), oriented. Each $\theta \in M$ defines a loss density

$$\lambda_\theta = \ell(x, \theta) \rho(x) \text{vol}_X \in \Omega^n(X),$$

where ρ is the density of the data with respect to the Riemannian volume of X . The population loss is $L(\theta) = \int_X \lambda_\theta = \langle \lambda_\theta, [X] \rangle$: the evaluation of a top-degree class against the fundamental class. The empirical loss replaces $[X]$ by the 0-chain $\frac{1}{N} \sum_i [x_i]$ — and the generalization gap is, quite literally, the error committed when evaluating a cocycle against a chain that only weakly approximates the fundamental cycle. This rereading is not vacuous: it converts questions about generalization into questions about *how well an empirical chain represents a homology class*, a problem with structure of its own (and with its own approximation theory: discrepancy, optimal transport, manifold sampling).

Poincaré duality takes the next step. On a compact oriented X of dimension n , the pairing

$$H_{\text{dR}}^k(X) \times H_{\text{dR}}^{n-k}(X) \longrightarrow \mathbb{R}, \quad ([\alpha], [\beta]) \mapsto \int_X \alpha \wedge \beta$$

is non-degenerate; equivalently, every cohomology class of degree k is dual to a cycle of codimension k . Suppose now that the loss is disaggregated: instead of a single scalar, we record the loss restricted to data submanifolds $C \subseteq X$ (subpopulations, strata, regimes):

$$L_C(\theta) = \int_C \iota_C^* \lambda_\theta^{(k)},$$

where $\lambda_\theta^{(k)} \in \Omega^k(X)$ is a loss density of degree adapted to $\dim C = k$. Model selection compared across families of subpopulations is then the study of the bilinear pairing

$$\mathcal{L} : H_{\text{dR}}^k(X) \times H_k(X) \longrightarrow \mathbb{R}$$

whenever the loss densities are closed (which happens, for instance, when $\lambda_\theta^{(k)}$ is built as the pull-back of volume forms of representation spaces along the feature map $\Phi_\theta : X \rightarrow \mathbb{R}^m$). In that regime a remarkable rigidity holds:

Proposition 3.1 (Homological rigidity of evaluation). *If $d\lambda_\theta^{(k)} = 0$, then $L_C(\theta)$ depends only on the class $[C] \in H_k(X; \mathbb{R})$. In particular: (a) the evaluated loss is invariant under deformations of the subpopulation C within its homology class — robustness by construction against perturbations of the evaluation cohort; (b) two models θ, θ' are indistinguishable by all evaluations $\{L_C\}_{[C] \in H_k}$ if and only if $[\lambda_\theta^{(k)}] = [\lambda_{\theta'}^{(k)}]$ in $H_{\text{dR}}^k(X)$, by the non-degeneracy of the Poincaré pairing.*

Item (b) suggests a notion of *cohomological equivalence of models*: the class $[\lambda_\theta] \in H_{\text{dR}}^\bullet(X)$ is the fingerprint of the model that stratified evaluations can see, and Poincaré duality guarantees that the evaluation panel $\{C\}$ is complete exactly when its classes generate $H_\bullet(X)$. A benchmark is, in this language, a basis of homology; a deficient benchmark is one whose classes fail to generate, and the failure modes it misses are precisely the annihilator of its span under the pairing. That this sounds like design of experiments is no accident: it is design of experiments in the correct category.

4 Conclusion: topological invariance against overfitting

Let us recapitulate the arc. The Fisher curvature produces, via Chern–Weil, cohomological invariants of the model space insensitive to parametrization and to the α -structure (§1). Degree-1 classes obstruct global potentials and endow the estimator with monodromy (§2). Losses, promoted to densities, turn evaluation into a Poincaré pairing between the model and the homology of the data (§3). The common thread is that *the cohomological is the stable*: that which survives deformations, reparametrizations, and perturbations of the evaluation chain.

Overfitting is, seen from here, the opposite phenomenon: fine dependence on the sample, on the seed, on the parametrization — dependence on representatives, not on classes. This motivates the final conjecture.

Conjecture 4.1 (Regularization by topological invariance). *Let $\mathcal{A}$ be an architecture with effective hypothesis space $M = \Theta/G$ and empirical loss L_N on a sample of size N . For c near the optimal value, let $b_k(N, c)$ be the Betti numbers of the sublevel set $M_{\leq c}^{(N)} = L_N^{-1}(-\infty, c]$, computed in a stable fashion (persistent homology over a range of c). Then:*

1. (**Topological stability $\Rightarrow$ generalization**) *If the persistence diagrams of $\{L_N\}$ are stable under resampling (bootstrap) of the sample — that is, if the topology of the sublevel sets near the optimum is an invariant of the distribution rather than of the sample — then the generalization gap is controlled by the bottleneck distance between the empirical and population diagrams.*
2. (**Cohomological stopping criterion**) *The onset of overfitting during training coincides with the appearance of short-persistence classes in $M_{\leq c}^{(N)}$ that have no counterpart in the population limit: the model begins to sculpt spurious topology dictated by noise. Penalizing the total persistence of such classes (a computable functional on the sampled landscape) is a consistent regularizer.*
3. (**Strong form**) *Among architectures of equal approximation capacity, those whose M has trivial reduced cohomology in the degrees accessible to the dynamics ($H_{\text{dR}}^1(M) = 0$ in particular: no period obstructions, no monodromy of the estimator) stochastically dominate, in generalization, those that do not.*

The conjecture is falsifiable piece by piece. Item 2 is directly experimental: the persistent homology of subsampled loss landscapes is computable in moderate dimensions and along random sections in high dimensions, and the correlation between spurious persistence and the generalization gap is measurable today. Item 3 connects with the literature on mode connectivity and network symmetries: explicitly quotienting by G (canonical parametrizations, neuron alignment) should, if the strong form is true, reduce the variance across runs and the generalization gap simultaneously. Item 1 is the deepest and the hardest: it asks for a stability theorem in the style of Cohen–Steiner–Edelsbrunner–Harer in which the noise perturbs not a fixed function but the measure that defines it.

If information geometry was the program of reading statistics through Riemann’s eyes, what is sketched here is the next program: reading it through de Rham’s. The Fisher metric measures; its characteristic classes *remember*. And perhaps that is the final lesson: a model that generalizes is a model that, like a cohomology class, does not depend on the representative through which the accident of the sample presented it.

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